



Review Test Submission: Assignment Week 5

User Mr W (Wouter) W Nel

Course EBN 122 S2 2017

Test Assignment Week 5

Started 8/28/17 8:52 PM

Submitted 8/28/17 10:59 PM

Due Date 8/28/17 11:59 PM

Status Completed

Attempt Score 14 out of 20 points

Time Elapsed 2 hours, 7 minutes

Instructions Read over the document *ClickUP Assignments Instructions*.

Before sending an enquiring regarding ClickUP question to any-one please verify that your problem is valid ... i.e. if you suspect incorrect marking first check with a fellow student or two.

Please direct any enquiries regarding ClickUP assignments to EBN2017.ClickUptest@gmail.com

Save the answer to each question as you complete it.

At the end of the assignment click Save and Submit.

Give your final correct answer 3 significant digits.

Make sure you use the FULL STOP for your decimal place marker.

You must indicate the units, unless otherwise specified:

- There must be NO space between the answer and the units
- There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V*
- Write out 'ohm' to indicate Ω (use singular)
- Use 'u' to indicate micro.
- Only use pico(p), nano(n), micro(u), milli(m), kilo(k), mega(M), giga(G) and tera(T); do not use centi, deci and deca

Results All Answers, Submitted Answers, Correct Answers, Feedback, Incorrectly Answered Questions

Question 1

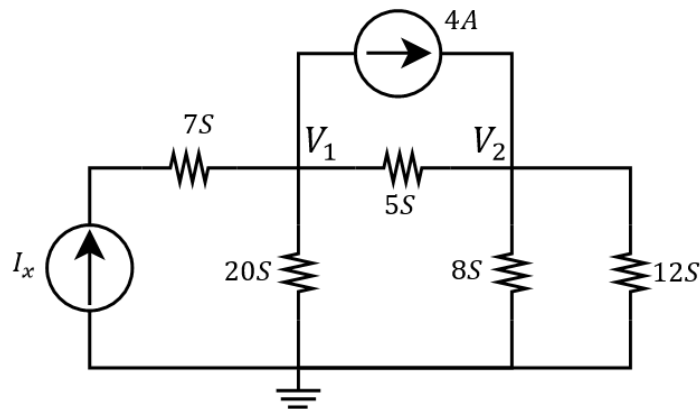
2 out of 2 points



Given $I_x = 5$, use inspection to set up a matrix of the following form in terms of nodal voltages:

$$\begin{bmatrix} 25 & -5 \\ -9 & 25 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} c \\ 4 \end{bmatrix}$$

Give the value of c.



Selected Answer: 1

Correct Answer: 1.00 ± 5%

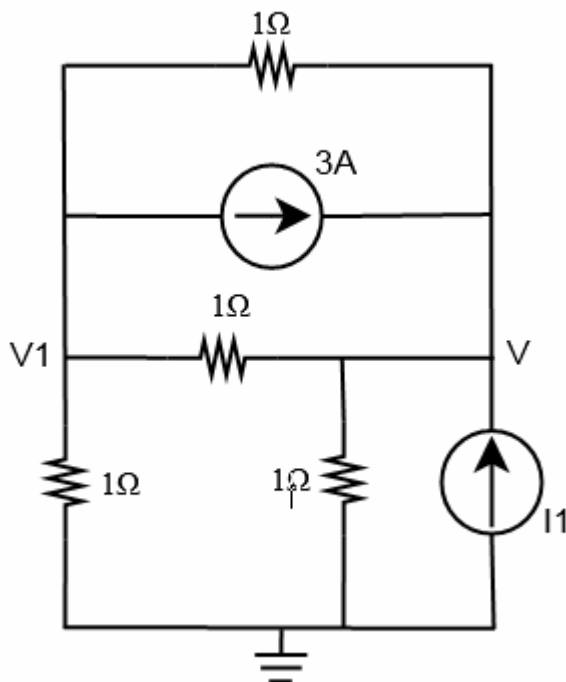
Question 2

2 out of 2 points



By inspection, write down the node-voltage equations for the circuit (in matrix form). If $I_1 = 11$, provide b:

$$\begin{bmatrix} 3 & a \\ -2 & 3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -3 \\ b \end{bmatrix}$$



Selected Answer: 14

Correct Answer: 14.00 ± 5%

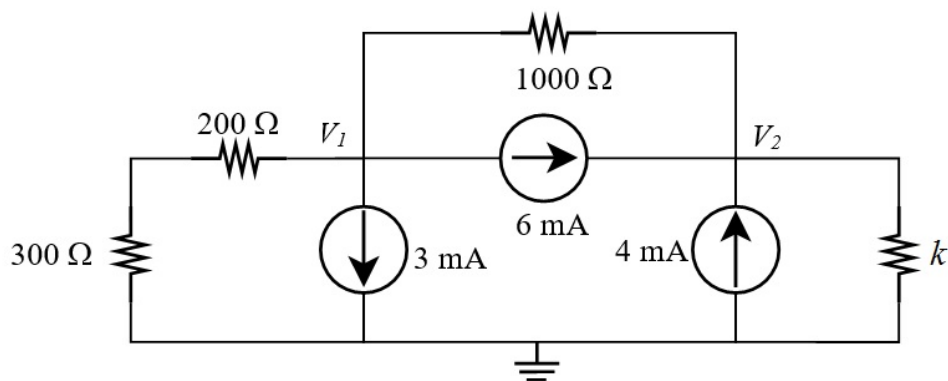
Question 3

2 out of 2 points



With $k = 3$ ohms, find P using inspection.

$$\begin{pmatrix} 0.003 & -0.001 \\ -0.001 & P \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} -0.009 \\ 0.01 \end{pmatrix}$$



Selected Answer: ☒ 0.334

Correct Answer: ☒ $0.33 \pm 5\%$

Question 4

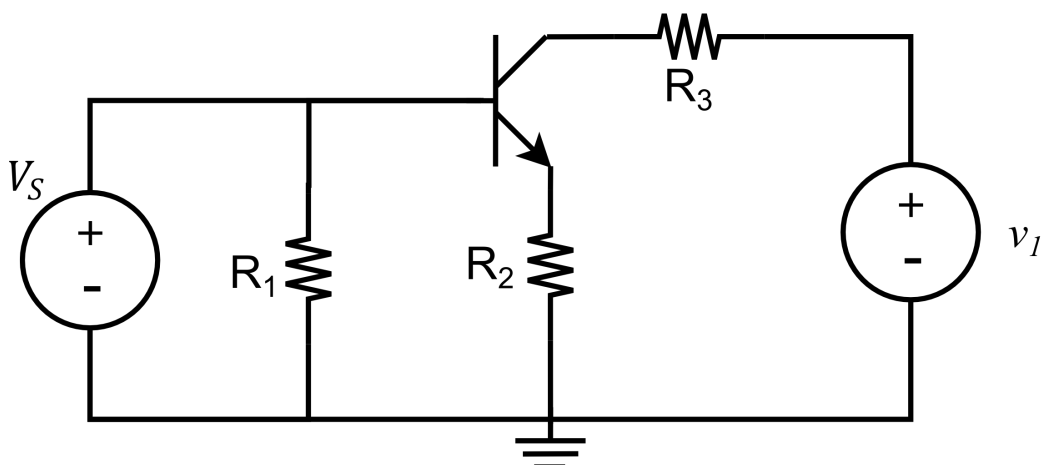
2 out of 2 points



Given $\beta=95$, $V_{BE} = 0.7V$ and $I_B = 2 \mu A$,

☒ $R_1 = 1k\Omega$, $R_2 = 4k\Omega$, $R_3 = 10k\Omega$ and $v_1=16V$

Determine V_{CE} .



Selected Answer: ☒ 13.3V

Correct Answer: ☒ $13.332 \pm 5\%$ (V)

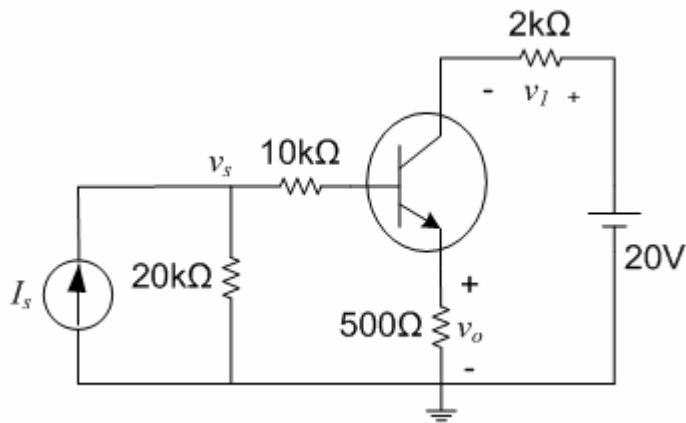
Question 5

0.5 out of 2 points



$\beta = 100$ and $V_{BE} = 0.7 V$

Determine V_0 if $I_s = 0.6 \text{ mA}$.



Selected Answer: ☒ 5.6V

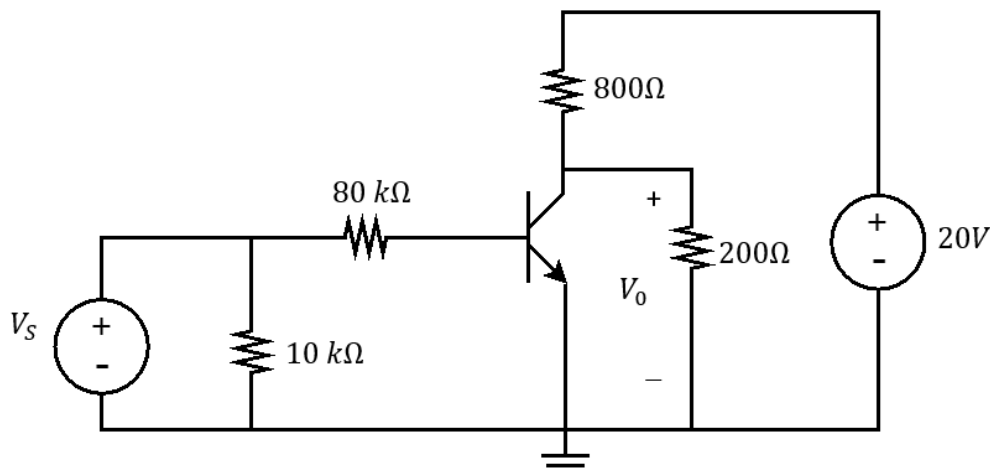
Correct Answer: ☒ $7.09 \pm 5\%$ (V)

Question 6

0.5 out of 2 points



Given: $\beta = 100$, $V_{BE} = 0.7\text{V}$ and $V_S = 13\text{V}$, find V_O .



Selected Answer: ☒ 5.48V

Correct Answer: ☒ $3.23 \pm 5\%$ (V)

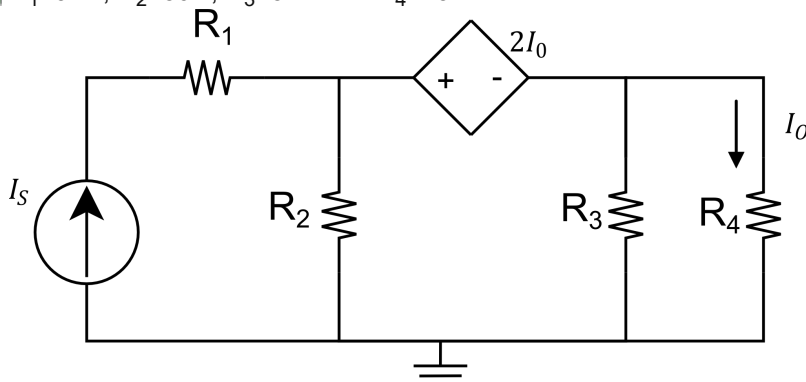
Question 7

2 out of 2 points



Using the principle of linearity, determine I_O/I_S

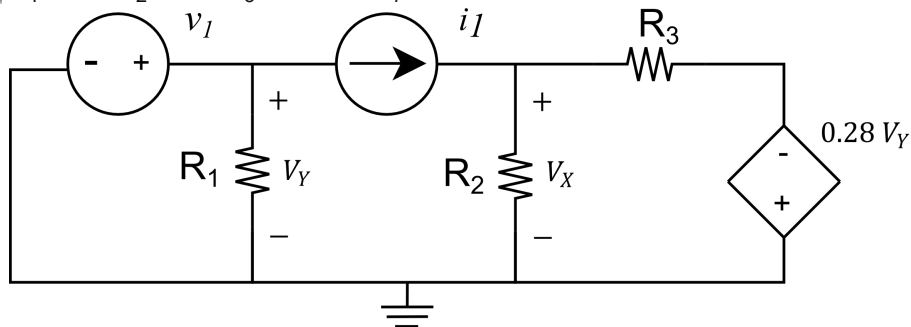
☒ $R_1=92\Omega$, $R_2=50\Omega$, $R_3=57\Omega$ and $R_4=73\Omega$



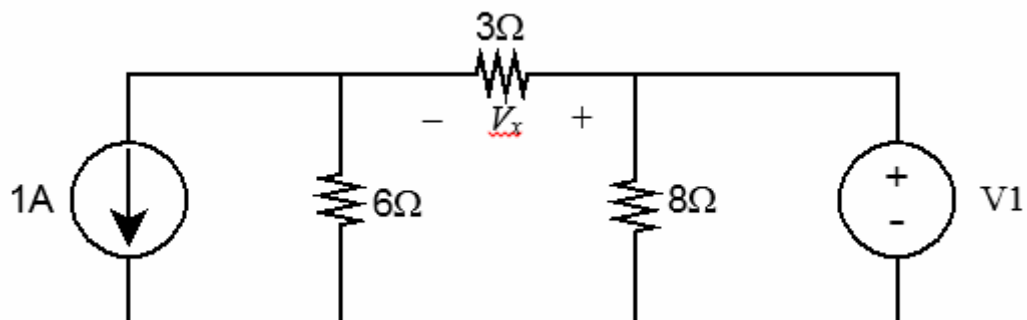
Selected Answer: ☒ 0.265

Correct Answer: ☒ $0.265 \pm 2\%$ **Question 8**

2 out of 2 points

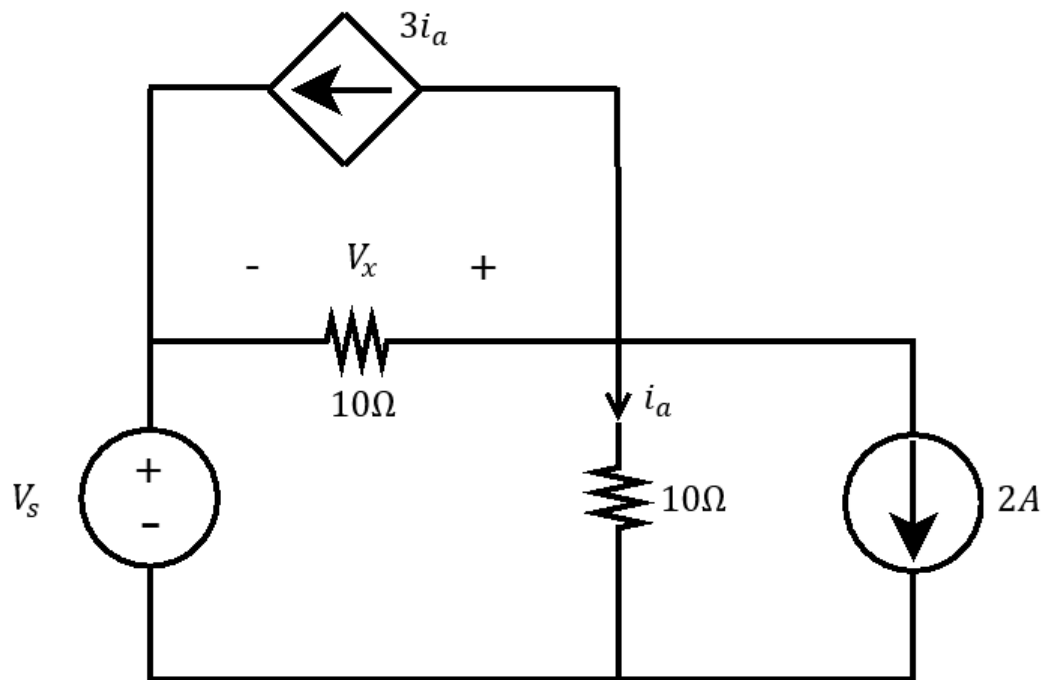
Determine the individual contribution of the v_I current source to V_X .☒ $R_1=19\Omega$, $R_2=73\Omega$, $R_3=45\Omega$ and $v_I=11V$ and $i_I=105mA$.Selected Answer: ☒ $-1.91V$ Correct Answer: ☒ $-1.905 \pm 5\%$ (V)**Question 9**

0.5 out of 2 points

Use superposition, if $V1 = 78 V$. Determine V_X only due to the voltage source.Selected Answer: ☒ $-26V$ Correct Answer: ☒ $26.00 \pm 5\%$ (V)**Question 10**

0.5 out of 2 points

Determine V_X due to the $V_S = 14$ voltage source only:



Selected Answer:  -18.7V

Correct Answer:  -11.20 ± 5% (V)

Monday, August 28, 2017 10:59:57 PM SAST

← OK